

FIRE DETECTOR ALARM

A report submitted in partial fulfillment of the Academic requirements.

SMART PRODUCT DESIGN

CSE/D2-32

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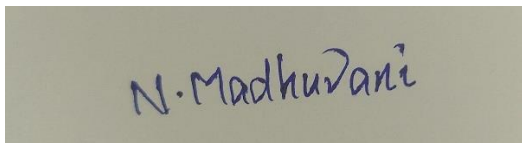
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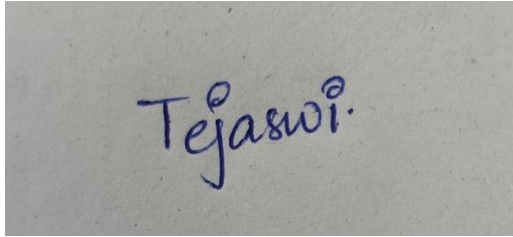
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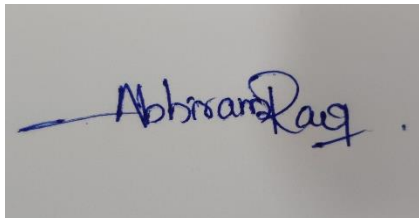
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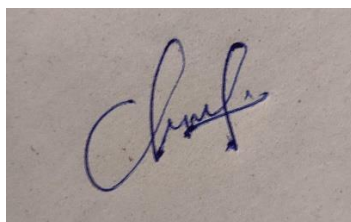
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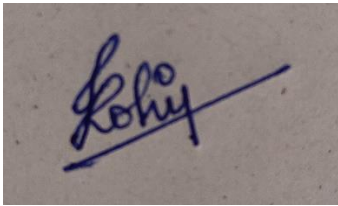
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CERTIFICATE

This is to certify that the course project entitled “FIRE DETECTOR ALARM” is the bonafide work carried N.Madhuvani (2303A52117), B.Tejaswi (2303A52123), T.Abhiram (2303A51426), K.Charan Yadav(2303A51614), A.Rohith (2303A52110) in the partial fulfilment of the requirement for the award of course Smart Product Design during the academic year 2023-2024 under our guidance and Supervision.

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ABSTRACT

FIRE DETECTOR ALARM is a safety device designed to prevent from fire accidents. Typically utilizing sensors to walls the buzzer activates when fire are detected, reducing the rate of risk of accidents, major and minor injuries for people. The implementation of fire detection systems is crucial for safeguarding lives and property against the devastating impact of fires. This abstract outlines the essential aspects involved in the deployment of fire detectors. It begins with the selection of appropriate sensors, such as smoke, heat, or flame detectors, tailored to the specific environmental conditions. The design process encompasses circuitry considerations, including analog signal processing or digital interfacing with microcontrollers, ensuring accurate detection and alarm triggering.

CHAPTER 1

INTRODUCTION

A flame detector is a critical component of fire safety systems, designed to quickly identify and alert people to the presence of smoke, heat, or flames. These devices play a crucial role in preventing the rapid escalation of fires, allowing for timely response and evacuation. Modern detectors use advanced technologies to sense different fire indicators and are integrated into comprehensive fire alarm systems for maximum effectiveness in safeguarding lives and property.

OBJECTIVES:

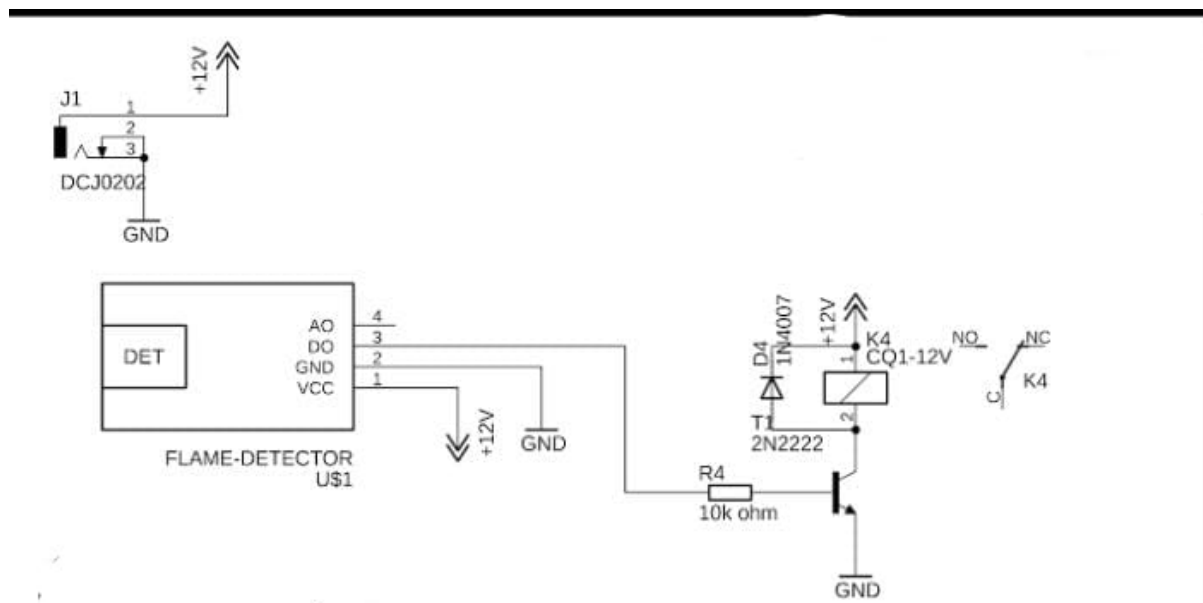
- * Understanding the basic Security and Flame detector system.
- * Understanding the Security system components.
- * Understanding zones and the different wiring configurations.
- * Understanding input and output devices.
- * Understanding system communications and devices.
- * Knowing what to have with you on site.

CHAPTER 2

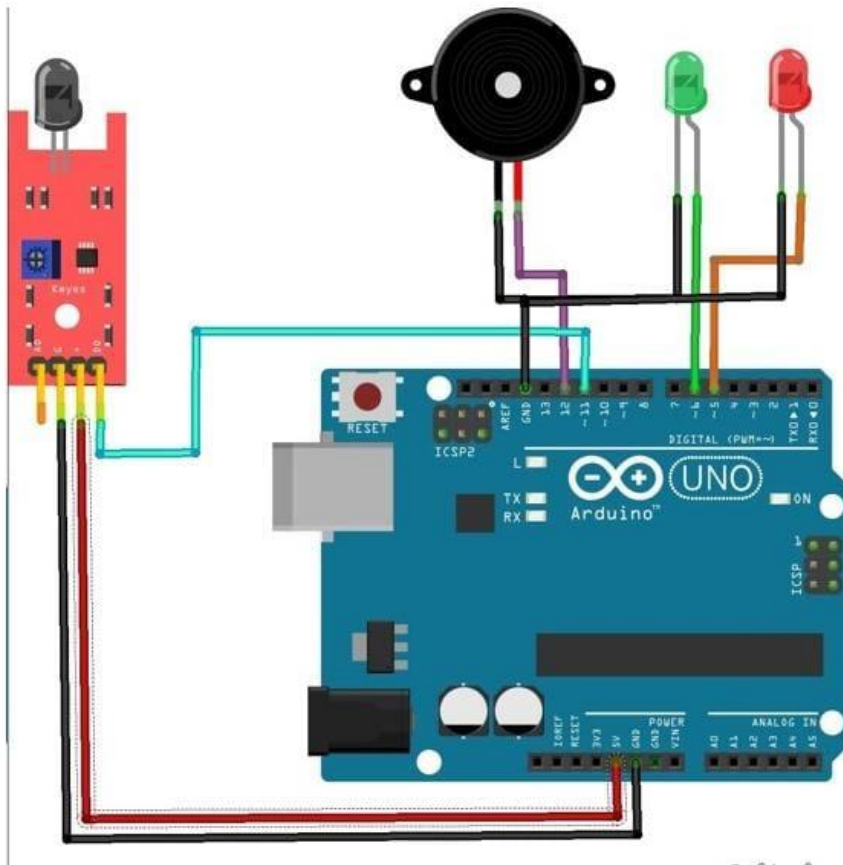
SYSTEM DESCRIPTION

Flame detector system is a building system designed to detect and alert occupants and emergency forces of the presence of smoke, fire,, or other fire-related emergencies. Fire alarm systems are required in most commercial buildings. They may include smoke detectors, heat detectors, and manual fire alarm activation devices, all of which are connected to a Fire Alarm Control Panel (FACP) normally found in an electrical room or panel room. Fire alarm systems generally use visual and audio signalization to warn the occupants of the building . Some fire alarm systems may also disable elevators, which under most circumstances, are unsafe to use during a fire. As the working of fire indicator using Arduino.

2.1 Block Diagram



2.2 Circuit Diagram

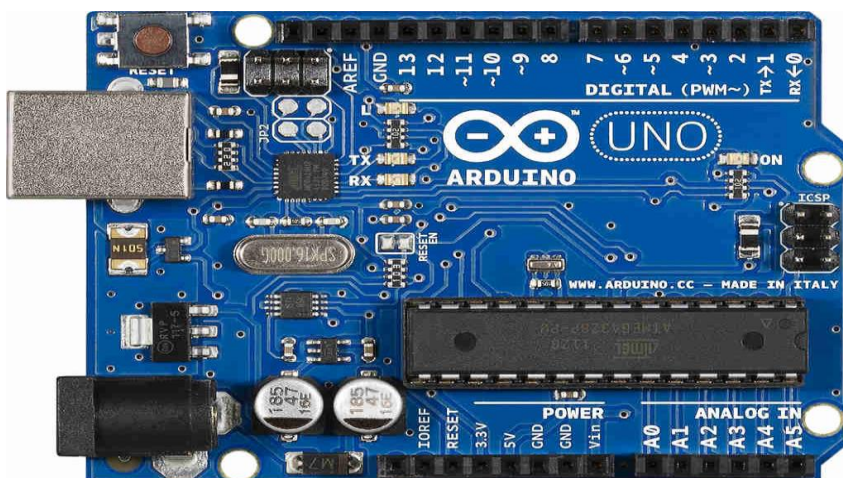


CHAPTER 3

HARDWARE/SOFTWARE TOOLS

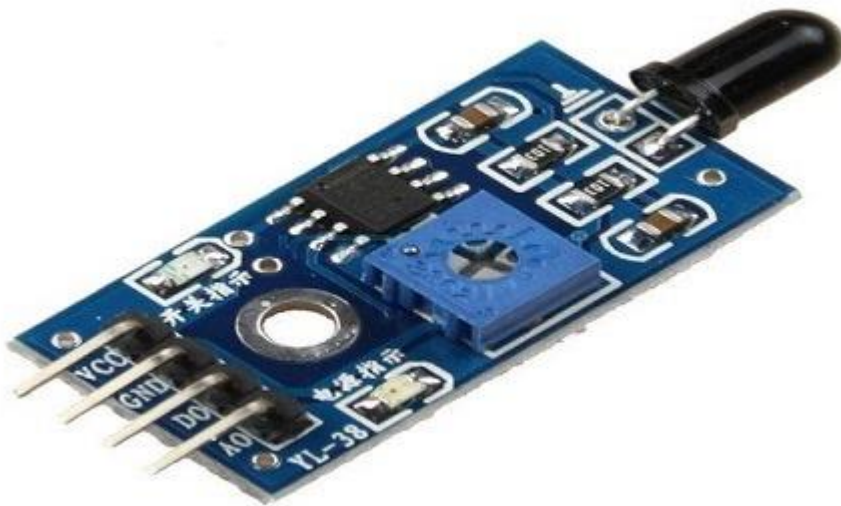
Arduino UNO:

The Arduino Uno is a microcontroller board based on the ATmega328 (datasheet). It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started. The Uno differs from all preceding boards in that it does not use the FTDI USB-to-serial driver chip. Instead, it features the Atmega16U2 (Atmega8U2 up to version R2) programmed as a USB-to-serial converter. Revision 2 of the Uno board has a resistor pulling the 8U2 HWB line to ground, making it easier to put into DFU mode. Revision 3 of the board has the following new features: 1.0 pinout: added SDA and SCL pins that are near to the AREF pin and two other new pins placed near to the RESET pin, the IOREF that allow the shields to adapt to the voltage provided from the board. In future, shields will be compatible both with the board that use the AVR, which operate with 5V and with the Arduino Due that operate with 3.3V. The second one is a not connected pin, that is reserved for future purposes. Stronger RESET circuit. Atmega 16U2 replace the 8U2. "Uno" means one in Italian and is named to mark the upcoming release of Arduino 1.0. The Uno and version 1.0 will be the reference versions of Arduino, moving forward. The Uno is the latest in a series of USB Arduino boards, and the reference model for the Arduino platform; for a comparison with previous versions, see the index of Arduino boards.



FLAME SENSOR :

The type of flame sensor in this fire alarm system that used is MQ-2. This type of flame sensor can detect flames. This flame sensor has 4 pins which are AO, DO, ground and VCC. FLAME SENSOR The flame sensor is to detect the presence of flame at the surrounding. This component has four pins which are pin AO, ground, VCC and DO. The pin DO is connected to the digital pin 2 at Arduino Uno while the ground and VCC are connected to ground pin and power 5Volt respectively. In this fire alarm system, the threshold is set 260. When the flame sensor value at pin DO is less than 260, the buzzer will activate.



BUZZER :

A buzzer is an electronic component that produces sound when an electric current is applied. It typically consists of a coil of wire and a diaphragm, and the sound is generated by the vibration of the diaphragm. In a circuit, a buzzer is often connected to a power source, and when the circuit is closed, the buzzer activates, producing an audible tone. Buzzer applications range from simple alert systems to more complex uses in electronic devices and alarms. They are commonly used in various electronic projects, such as timers, doorbells, and security systems.



USB Cable:

A USB (Universal Serial Bus) cable is a standardized cable that is widely used for connecting and transferring data between electronic devices. Here are some key points about USB cables. USB has gone through several versions, each offering improvements in speed and capabilities. Common versions include USB 1.0, USB 2.0, USB 3.0, USB 3.1, and USB 3.2. USB cables are commonly used to transfer data between devices such as computers, smartphones, cameras, printers, and more. The data transfer speed depends on the USB version.



Connecting Wires:

Connecting wires in the smart blind stick are crucial for integrating various sensors and components, such as obstacle detectors, distance sensors, and feedback mechanisms. These wires enable communication between different parts of the device, facilitating the seamless operation of its functionalities. Properly organized and insulated wires ensure reliable data transmission and power distribution, optimizing the stick's performance. Additionally, efficient wire management minimizes the risk of entanglement and damage, enhancing the stick's durability and user experience. Overall, the careful connection of wires is essential for the functionality, reliability, and safety of the smart blind stick, empowering visually impaired individuals in their daily navigation.



CODE:

```
int sensorPin=A0;
int buzzer=13;
int G_led=8;
int R_led=9;
int read_value;
void setup(){
  pinMode(sensorPin,INPUT);
  pinMode(buzzer,OUTPUT);
  pinMode(R_led,OUTPUT);
  pinMode(G_led,OUTPUT);
}
void loop(){
  read_value=digitalRead(sensorPin);
  if (read_value==1){
    digitalWrite(buzzer,LOW);
    digitalWrite(R_led,LOW);
    digitalWrite(G_led,HIGH);
  }
```

```
else{  
    digitalWrite(buzzer,HIGH);  
    digitalWrite(R_led,HIGH);  
    digitalWrite(G_led,LOW);  
    delay(1000);  
}  
delay(100);  
}
```

CHAPTER 4

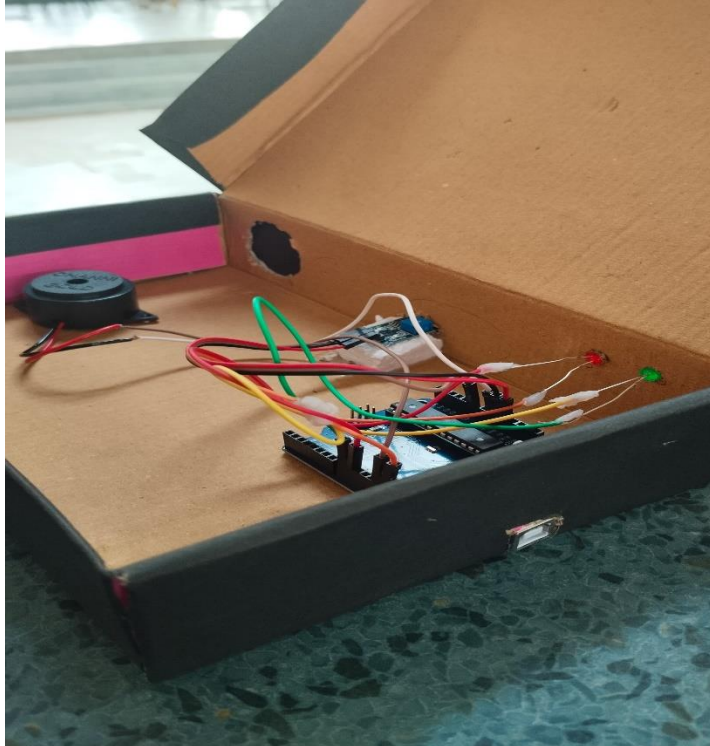
IMPLEMENTATION

Implementing a fire detector involves a multifaceted process that starts with selecting the appropriate sensor for your specific needs and environment. Whether it's a smoke detector, heat detector, or flame detector, the choice must align with the characteristics of the space it's intended to monitor. Once selected, designing the circuit comes next, which may vary depending on whether you're using analog or digital sensors. Analog sensors like smoke or heat detectors require circuitry for signal amplification, filtering, and conditioning, while digital sensors typically interface with microcontrollers like Arduino or Raspberry Pi. These microcontrollers process sensor data and execute programmed instructions to trigger alarms or notifications when fire hazards are detected. Power supply considerations are paramount, ensuring continuous operation even during emergencies. Integration with existing alarm systems or designing standalone systems with alarm outputs is essential for effective fire detection. Rigorous testing, calibration, proper installation, and ongoing maintenance are vital steps to ensure the system's reliability and compliance with safety standards.

CHAPTER 5

RESULTS

Managed to successfully apply the FLAME DETECTOR SYSTEM using Arduino and it was user friendly and cost effective. User friendly as in anyone can use just a click of a button on an android screen and everything works. And it is cost effective as in it will cost exactly as the project requires (optimum price).





- A. **PROBLEM:** What overall problem(s) does the proposed invention solve or what purpose does it serve? (Note: Please be specific, spell out acronyms and provide enough layman level detail to fully explain the problem.)

A fire detector alarm aims to solve is the early detection and warning of fires in order to protect lives and property. These devices serve the critical purpose of detecting the presence of smoke, heat, or flames, which are indicators of a fire outbreak. By promptly alerting occupants of a building or space to the presence of a fire, these alarms provide crucial time for evacuation and for initiating firefighting measures to minimize damage. In simpler terms, imagine a fire breaking out in a home or a building when no one is aware. Without a fire detector alarm, people wouldn't know about the fire until they see or smell smoke, or worse, until the fire spreads and causes significant damage. But with a fire detector alarm in place, as soon as smoke or heat is detected, the alarm sounds loudly, alerting everyone in the vicinity to the danger, allowing them to escape to safety and enabling firefighters to respond quickly to extinguish the fire.

So, the problem these alarms solve is essentially the lack of early warning in the event of a fire, providing a vital layer of protection for people and property.

B. EXISTING SOLUTIONS / PRIOR ART/RELATED APPLICATIONS & PATENTS:

1. List any known products, or combination of products, currently available to solve the same problem(s). What is the present commercial practice?

Special Features

1. Smoke Alarms: These detectors are designed to sense smoke particles in the air. They are typically installed in residential and commercial buildings and are often battery-powered or hardwired into the building's electrical system.
2. Heat Detectors: Heat detectors activate when they sense a significant rise in temperature, indicating the presence of a fire. They are commonly used in areas where smoke detectors might not be suitable, such as kitchens or garages.
3. Combination Smoke and Carbon Monoxide Alarms: These alarms provide dual functionality by detecting both smoke and carbon monoxide (CO) in the air. CO is a colorless, odorless gas produced by incomplete combustion, and it poses a significant health risk.
4. Smart Fire Detectors: These detectors are connected to a network and can send alerts to smartphones or other devices when smoke or fire is detected. They may also offer additional features such as integration with smart home systems and remote monitoring capabilities.
5. Flame Detectors: These detectors are designed to sense the presence of flames, making them suitable for environments where smoke or heat detectors may not be effective, such as outdoor areas or spaces with high levels of dust or steam.

2. In what way(s) do the presently available solutions fall short of fully solving the problem?

1. False Alarms: One common issue with traditional smoke alarms is the occurrence of false alarms, triggered by cooking smoke, steam, or dust. These false alarms can lead to complacency among occupants and may result in the disconnection or disabling of the alarm, compromising safety.
2. Limited Coverage: Traditional smoke alarms are typically installed on ceilings, which may not provide adequate coverage in all areas of a building, especially in large or complex structures with multiple levels or rooms.
3. Lack of Integration: While some smart fire detection systems offer integration with other smart home devices, the level of integration with building management systems or emergency response systems may still be limited. This can hinder the efficiency of emergency response and evacuation procedures.
4. Maintenance Requirements: Fire detection and alarm systems require regular maintenance, including testing, cleaning, and battery replacement, to ensure their reliability. However, these maintenance tasks may be overlooked or neglected, leading to reduced effectiveness during a fire emergency.

3. Conduct key word searches using Google and list relevant prior art material found?
 1. "Smoke alarm technology advancements": This search yields results discussing recent advancements in smoke alarm technology, such as improved sensors, connectivity options, and integration with smart home systems.
 2. "Fire detection system patents": Searching for patents related to fire detection systems can uncover a wealth of prior art material, including detailed descriptions of innovative technologies and designs.
 3. "Heat detector false alarm mitigation": This search focuses on strategies and technologies aimed at reducing false alarms triggered by heat detectors, such as improved algorithms or sensor calibration methods.
 4. "Smart fire detection devices": Exploring the latest developments in smart fire detection devices can reveal prior art related to features like wireless connectivity, remote monitoring, and predictive analytics.

C. DESCRIPTION OF PROPOSED INVENTION:

How does your idea solve the problem defined above? Please include details about how your idea is implemented and how it works?

D. NOVELTY: Please provide a **one-sentence** description of what distinguishes your idea from the prior art. This is a statement of what is new, and not a business case.

The novel aspect of this fire detector alarm lies in its integration of advanced machine learning algorithms, enabling real-time analysis of environmental factors to differentiate between genuine fire threats and false alarms with unprecedented accuracy.

E. COMPARISON: Please provide advantages and basic differences of the proposed solution over previous solutions.

ADVANTAGES:

- *Early warning
- *Properly protection
- *Life safety
- *Legal and insurance requirments
- *Easy and Affordable

DIFFERENCES:

- *small in size
- *low cost
- *easy to install

CHAPTER 6

CONCLUSION

In conclusion, flame detectors play a crucial role in ensuring safety by swiftly identifying and alerting to the presence of flames. With their ability to detect fires in various environments, these devices contribute significantly to early fire detection and prevention, ultimately safeguarding lives and property. As technology continues to advance, the ongoing improvement of flame detection systems remains essential for enhanced reliability and responsiveness in fire safety measures.

The future of flame detectors involves integration with IoT, smart building systems, c wireless communication, increased sensitivity, and enhanced energy efficiency. These developments aim to provide more reliable and sophisticated fire detection for heightened safety and protection.

CHAPTER 7

REFERENCES

We referred this project from Youtube and google. We collected some information of components from google sites.

NFPA 72, developed by the National Fire Protection Association, offers comprehensive standards for fire alarm systems.

Underwriters Laboratories (UL) certifies fire alarm products and ensures their compliance with safety standards. The International Building Code (IBC) includes regulations pertaining to fire detection and alarm systems in buildings. Manufacturer documentation and engineering journals also provide valuable insights into fire detector alarm technology and advancements.